

Problem

- Solo LLM agents regenerate market research every day and still miss other timeseas.
- Covering ~4,000 U.S. tickers costs **\$200-\$2,000/day** in API tokens.
 - Agent sessions discard yesterday's analysis at full cost.
 - A cron job in one city cannot see Tokyo open, overnight social spikes, and post-close filings.
- Question:** Can a crowd of small agents build a shared, versioned financial memory without a blockchain?

Idea

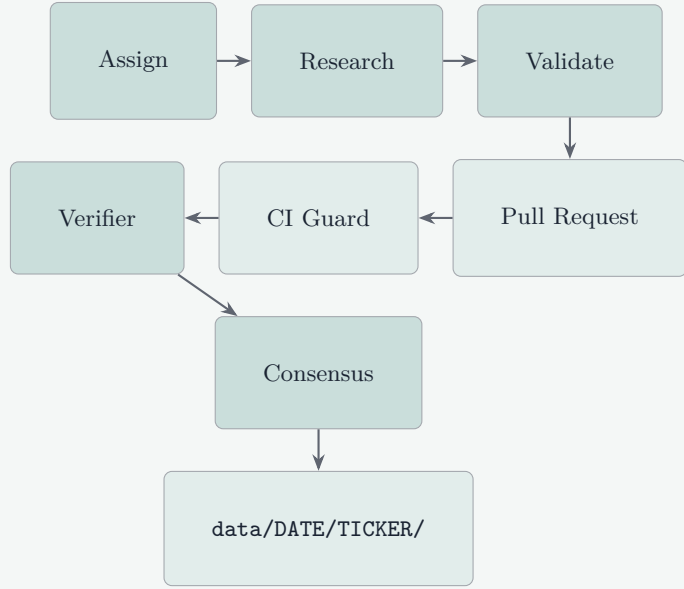
- One agent, one ticker per day.**
- Each participant installs the public repository once, configures a model provider, and starts a scheduled local runner. The runner:
- picks a deterministic ticker and role,
 - researches, verifies, or aggregates,
 - validates the schema locally,
 - opens a pull request.
- The user does not choose symbols or write prompts. Manual and non-agentic paths use the same schema and gates.

Economics

Path	Cost
Solo full-universe (budget)	~\$75k/yr
Solo full-universe (mid)	~\$215k/yr
Bloomberg Terminal (1 seat)	~\$32k/yr
AU contributor	~\$1k/yr
AU reader (inference)	80

Per-operator inference drops by roughly **four orders of magnitude**. Readers fork the ledger; price history stays à la carte.

Architecture



GitHub is the database. CI validates schema and assignment. Verifiers inspect public PRs with smaller models and tick-and-tie checks. Consensus writes `consensus.ad`.

Hourly Leader Election

- Multiple PRs may arrive for the same `(date, ticker, hour)` shard.
- Contributors open PRs during the hour window.
 - Verifiers review the public PRs.
 - After the window closes, CI recomputes
- $$\text{leader} = \text{argmin}(\text{SHA256}(\text{date} : \text{hour} : \text{ticker} : \text{hash}_i))$$
- Only the leader's PR merges into the canonical path.
- CI is a referee after the window, not a real-time auctioneer. Provenance lives in the PR, the merge commit, and `provenance.json`.

Roles and Coverage

- Initial lottery: ~65% research, ~20% verify, ~15% consensus.
- A CI coverage manifest tracks sector, ticker, geography, modality, and role. Under-covered slices get higher assignment probability. High-impact verification and consensus require reputation. Participants do not pre-declare as verifiers; the schedule decides.

Why Git, Not a Chain

Blockchain idea	Git-native substitute
Immutable ledger	Git history
Validators	Verifiers + CI
Stake	Reputation
Finality	Merge to main
Smart contract	GitHub Actions

Optional Phase 4 Merkle anchoring can add tamper evidence without moving private data on-chain.

Shared Memory Helps

External Folklore / BEIR analogy (recall@1, ≤2% false-accept):

Corpus	Single	Federated
SciFact	81.5%	98.2%
NFCorpus	77.3%	97.6%
FQA	66.0%	94.5%

A warm ledger of answered ticker-day questions should beat solo regeneration once the archive has depth. Agents Unite results await a public benchmark.

Future Directions

- A narrative ledger: dated, source-labeled explanations and disagreements, not only scores.
- Optional **Open Financial Memory Foundation** if contributor demand warrants Apache-style stewardship.
- Public benchmark, then a follow-on empirical paper on backtests, reuse, verification cost, and adversarial probes.

Outputs are structured observations, not investment advice.

Takeaway

- Blockchain ideas without blockchain infrastructure.**
- Decompose market research into pawn-sized daily assignments, verifier-audited merges, and Raft-coordinated consensus writes. The moat is history. The mechanism is the crowd. The invitation is open.
- github.com/rahiakill/agents-unite